

OsdagBridge Functional Testing & Issue Reporting

FOSSEE Screening Task Submission

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Introduction

This report documents the functional testing I conducted on the OsdagBridge software as part of the FOSSEE screening task. Testing covered input validation, custom property handling, multi-run stability, 3D CAD/graphical output verification, and menu/UI functionality. Both working and failed test scenarios are documented below, followed by a summary of GitHub issues raised against the repository with suggested code-level fixes.

1 Test Case Log

1.1 Test 1: Structure Type Dropdown

- **Action:** Checked the available options in the Structure Type dropdown.
- **Observation:** Only “Highway Bridge” is selectable. The other available option is greyed out, with no tooltip or explanation provided for why it is disabled.
- **Status:** Minor issue (Low severity)

1.2 Test 2: Span Field Decimal Precision

- **Action:** Entered a long decimal value for Span, e.g. 30.123456789.
- **Observation:** The application silently rounded the value to 30.12 without informing the user that the value had been changed, resulting in a loss of precision without user awareness.
- **Status:** Minor issue (Low severity)

1.3 Test 3: Graphics / Database / Help Menus

- **Action:** Clicked each of the Graphics, Database, and Help menus in the top menu bar, followed by clicking each individual option within the opened dropdowns.
- **Observation:** All dropdowns open and close correctly, but no action is triggered when clicking any option under any of the three menus.
- **Root Cause (from source inspection):** All `QAction` objects for these menus are created and added to their respective menus, but none are connected via `.triggered.connect()` to a handler function – so the UI renders correctly but no click behavior is wired up.
- **Status:** Bug (Medium severity) – Filed as [GitHub Issue #53](#)

1.4 Test 4: Custom Material / Geometric Properties

- **Action:** Entered a custom value for Number of Girders = 4 (instead of default).
- **Observation:** The application automatically recalculated and updated the Girder Spacing to 10.97 m. The 3D model updated accordingly, and the value matched the summary panel exactly.
- **Status:** Working as expected (Pass)

1.5 Test 5: Multi-Run Stability

Three consecutive design runs were performed by unlocking the input panel, modifying parameters, and re-running the Design function each time.

Run	Action / Parameters Changed	Result
Run 1	Span = 23.0 m, clicked Design	3D view updated correctly and matched the summary panel.
Run 2	Span changed to 30.0 m, Carriageway Width changed to 17 m	3D view updated correctly and matched the summary panel. No stale data observed from previous run.
Run 3	Skew Angle changed to 10.0°, Span changed to 40.0 m	3D view updated correctly and matched the summary panel. Application remained stable across all three sequential runs with no crashes.

Overall Observation: The 3D CAD view and the summary panel (Overall Bridge Width, Span Length, Girder Spacing, Deck Thickness) updated consistently and correctly across all three consecutive runs, with no evidence of stale data or instability.

1.6 Test 6: Input Panel Lock After Design

- **Action:** After clicking Design, attempted to scroll up/down within the (now locked) input panel to review previously entered values.
- **Observation:** Once Design is run, the input panel locks (as expected, to prevent edits without re-triggering a redesign). However, scrolling within the locked panel does not work either – there is no way to scroll up or down to review the variety of inputs that were previously entered without first unlocking the panel.
- **Status:** Usability issue (Low–Medium severity) – Users cannot review their full set of entered inputs while the design is locked, which reduces confidence in verifying what was actually submitted for the current design run.

2 UI Inputs vs. Input/Output Dictionary Comparison

- Values entered via the Basic Inputs and Additional Inputs panels (e.g. Span, Carriageway Width, Number of Girders, Skew Angle) were cross-checked against the values reflected in the generated 3D model summary panel (Overall Bridge Width, Span Length, Girder Spacing, Deck Thickness).

- In all tested cases, the UI-entered values matched the corresponding output values shown in the summary panel and 3D model exactly, indicating consistent data flow between the input layer and the design/output layer for the parameters tested.

3 Summary of GitHub Issues Filed

Issue #	Title	Severity	Link
53	Graphics, Database, and Help menu dropdowns open but options are unresponsive	Medium	View Issue

Conclusion

Testing of OsdagBridge covered input validation, custom property handling, multi-run design stability, and menu/UI functionality. The core Basic Inputs → Design → 3D CAD workflow was found to be stable and consistent across repeated runs with varying parameters. Several minor UI/UX issues were identified (dropdown option clarity, silent decimal rounding, locked-panel scroll limitation), along with one confirmed functional bug affecting the Graphics, Database, and Help menus, which has been filed on the project repository with a suggested code-level fix.